Homework #7

This is the homework for week 9. The due date is 11:59 p.m., November 12. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (30 points) Assume that IQ scores for a certain population are approximately $N(\mu, 100)$. To test H_0: $\mu = 110$ against the one-sided alternative hypothesis $H_1 : \mu > 110$, we take a random sample of size $n = 16$ from this population and observe $\bar{x}_n = 113.5$.

(a) Do we retain or reject H_0 at the 5% level/size?

(b) Do we retain or reject H_0 at the 10% level/size?

(c) Derive the value of p value.

(**Question2):** (20 points) Let X equal the forced vital capacity (FVC) in liters for a female college student. (The FVC is the amount of air that a student can force out of her lungs.) Assume that the distribution of X is approximately $N(\mu, \sigma^2)$. Suppose it is known that $\mu = 3.4$ liters. A volleyball coach claims that the FVC of volleyball players is greater than 3.4. She plans to test her claim with a random sample of size $n = 9$.

(a) Define the null hypothesis.

(b) Define the alternative (coach's) hypothesis.

(c) Define the test statistic.

(d) Define a critical region for which size $\alpha = 0.05$. Draw a figure illustrating your critical region.

(**Question3):** (30 points) Let X and Y denote the weights in grams of male and female common gallinules, respectively. Assume that X is $N(\mu_X, \sigma_X^2)$ and Y is $N(\mu_Y, \sigma_Y^2)$.

(a) Given $n_X = 16$ observations of X and $n_Y = 13$ observations of Y, define a test statistic and a critical region for testing the null hypothesis $H_0 : \mu_X = \mu_Y$ against the one-sided alternative hypothesis $H_1 : \mu_X > \mu_Y$. Let size $\alpha = 0.01$. (Assume that the variances are equal.)

(b) Given that $\bar{x}_n = 415.16$, $s_X^2 = 1356.75$, $\bar{y}_n = 347.40$, and $s_Y^2 = 692.21$, calculate the value of the test statistic and state your conclusion.

(c) Although we assumed that $\sigma_X^2 = \sigma_Y^2$, let us say we suspect that that equality is not valid. Thus, use the test proposed by Welch.

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

Continued with Question 2, the random sample yielded the following FVCs:

3.4, 3.6, 3.8, 3.3, 3.4, 3.5, 3.7, 3.6, 3.7.

(1) Calculate the value of the test statistic

(2) What is your conclusion?

(3) What is the approximate p value of this test? You need to derive the number of the P–value.